

Mantis IP Visualization: RAG Retrieval Pipelines

Standard retrieval, IPC-weighted scoring, and long-document chunking

Mantis IP Visualization

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Contents

1	Introduction	3
I	standard-rag/: baseline retrieval comparison	3
2	Goal	3
3	Setup and reused signal	4
4	Results	4
II	code-weight/: IPC-weighted retrieval	5
5	Goal	5
6	IPC taxonomy and scoring	6
7	Fusion with semantic score	6
8	Validation: clean vs. dirty IPC inputs	7
III	longdoc-rag/: chunked retrieval for long documents	7
9	Goal	7
10	Chunking methodology	8

11 Aggregating chunk scores into one ranking	8
12 End-to-end pipeline	9

1 Introduction

This document covers the retrieval tooling I build using The Lens Patent Corpus sample dataset (`lens-db-translated.json`, roughly 7,400 English-language patent records after filtering for non-empty title and abstract fields). The goal with these methods was to take a given query document/input and return a ranked list of the most relevant patents in the corpus. The core problem is that these inputs are diverse in both size and type (e.g. some inputs contain IPC codes, some are 1,000+ words, etc.). As such, there are 3 different pipelines in this repository. The first tests multiple standard RAG systems, the second incorporates IPC-classifying codes into the final document relevancy score. The third one uses document chunking to handle documents too long for one pass. All of the code referenced here is in the `rag-testing/` directory of the project GitHub repository.

Part I

standard-rag/: baseline retrieval comparison

2 Goal

Which retrieval method gives the best relevance per millisecond over the patent corpus? To answer this, I built three retrieval systems and tested them on the same nine test queries. These queries included patent abstracts, product descriptions of different lengths, and plain-language queries of different lengths.

- **BM25** is a classic sparse lexical retrieval method. It scores documents using term-frequency and inverse-document-frequency statistics over tokenized title and abstract text. It does not use an embedding model.
- **MiniLM** is a dense retrieval method. It uses `all-MiniLM-L6-v2`, a small general-purpose sentence-embedding model.
- **BGE-small** is a dense retrieval method. It uses `BAAI/bge-small-en-v1.5`, a model built specifically for retrieval. It adds an instruction prefix ("**Represent this sentence for searching relevant passages:** ") to the query text only, not to the corpus text.

For both dense methods, I L2-normalize the corpus embeddings when I build the index. This means cosine similarity becomes a simple dot product at query time, which is faster to compute. This is also why the embeddings cache (`*_embeddings.npy`) stores normalized vectors instead of raw ones.

3 Setup and reused signal

All three systems index the same text for each patent: the title and abstract combined. I embed or tokenize this text once and cache the result to disk (`results/bm25/`, `results/minilm/`, and `results/bge/`). This means repeat queries do not need to rebuild the index. The query times reported below measure only the score computation step. They do not include model or index load time.

4 Results

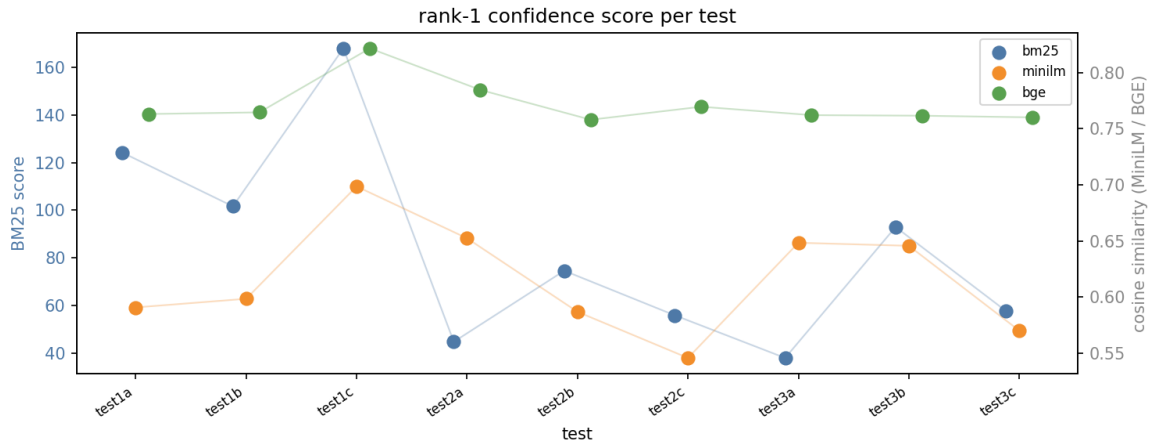


Figure 1: Rank-1 confidence score per test query. BM25 scores (left axis) are unbounded term-frequency scores. MiniLM and BGE scores (right axis) are cosine similarities in $[0, 1]$. BGE’s top score stays in the 0.75 to 0.82 range across every test. It is the flattest of the three lines, so it is the least sensitive to changes in query phrasing or length.

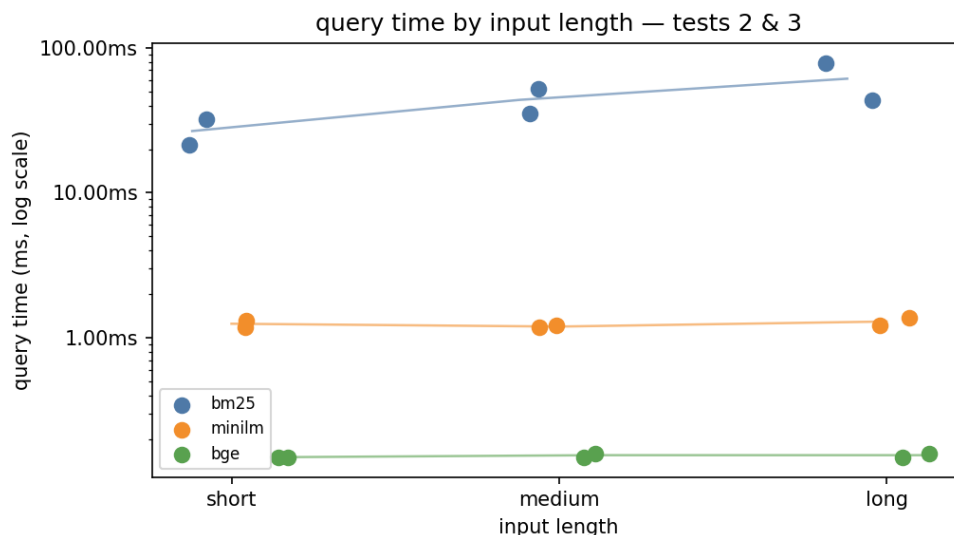


Figure 2: Query time by input length (log scale), tests 2 and 3. BGE is roughly two orders of magnitude faster per query than BM25 and stays flat as input length grows, since the entire query is reduced to one embedding regardless of length.

Metric	BM25	MiniLM	BGE-small
Query time (typical)	20 to 80 ms	1 to 3 ms	0.15 to 0.17 ms
Rank-1 score (mean, tests 1 to 3)	about 87 (unbounded)	about 0.62	about 0.78
Query-time scaling with input length	grows (roughly linear)	flat	flat
Short-vs-long overlap, Jaccard (top-10)	0.25 to 0.43	0.33 to 0.43	0.25

Takeaway: I chose BGE-small as the retrieval backbone for the two downstream pipelines, `code-weight/` and `longdoc-rag/`. It has the fastest query time of the three methods by a wide margin. It also has the highest and most stable rank-1 confidence scores. On top of that, it is a dense method with normalized embeddings, so it fuses cleanly with additional signal. That fusion is exactly what the IPC-weighting pipeline below needs.

Part II

code-weight/: IPC-weighted retrieval

5 Goal

Patent records carry IPC (International Patent Classification) codes. These codes form a hierarchical taxonomy assigned by patent examiners, and they are a human-curated relevance signal that is arguably stronger than free text alone. A pure text-embedding retrieval system throws this signal away entirely. My goal with this pipeline is to combine semantic text

similarity with IPC-taxonomy similarity. Two patents in the same narrow classification should rank higher relative to each other than raw cosine similarity alone would suggest. At the same time, the system should still fall back to plain semantic search when IPC codes are missing or not useful.

6 IPC taxonomy and scoring

An IPC code such as C07H21/04 breaks down into five hierarchical levels. Each level is more specific than the one before it:

Level	Example	Meaning
Section	C	Top-level domain (Chemistry and Metallurgy)
Class	C07	Organic chemistry
Subclass	C07H	Sugars, nucleosides, nucleotides
Main group	C07H21	Compounds containing nucleoside/nucleotide units
Subgroup	C07H21/04	Full code, specifically deoxyribonucleic acids

For a pair of IPC codes, I score their similarity based on the **deepest level at which they agree**:

Deepest matching level	Score
Subgroup (full code match)	1.0
Main group	0.8
Subclass	0.5
Class	0.2
Section only, or no match	0.0

A query usually carries several IPC codes, and so does each corpus document. To score a document, I look at every pair of a query code and a document code, and I take the **maximum** score across all of those pairs. This means a document only needs one strong classification match with the query to score well. It does not matter if its other codes are unrelated.

7 Fusion with semantic score

Before combining the two scores, I first min-max normalize the BGE cosine similarity across the whole corpus for the current query. This puts it in the same $[0, 1]$ range as the IPC score. Without this step, the IPC term would unfairly dominate, since cosine scores are usually compressed into a narrow band between 0.55 and 0.85, while IPC scores are spread across the full range. The final score is a weighted sum of the two:

$$\text{Score}_{\text{final}} = \alpha \cdot \text{cos}_{\text{norm}} + (1 - \alpha) \cdot \text{IPC}_{\text{score}}$$

I use $\alpha = 0.7$. This gives 70 percent weight to the semantic score and 30 percent weight to the IPC score. If a document has no IPC codes at all, I drop the IPC term and the score falls back to normalized cosine alone. This way, patents without IPC codes are never unfairly penalized.

8 Validation: clean vs. dirty IPC inputs

I wanted to confirm that the IPC term was actually moving rankings, rather than just sitting there without doing anything. To test this, I ran every test query twice while keeping the query text fixed. The first run used the real, semantically relevant IPC codes (`inputs-clean/`). The second run used IPC codes sampled from an unrelated technology area instead (`inputs-dirty/`).

For the biotech test query (`test1a`), the clean run used codes `A61K9/51`, `A61K48/00`, and `A61P35/00`. These codes cover drug delivery, gene therapy, and antineoplastic activity. The dirty run swapped in unrelated codes from automotive, construction, and mechanical domains instead (`B62D25/08`, `E04B1/00`, and `F16H57/00`). Here is the effect on the top result:

Variant	Rank-1 doc	cos_norm	IPC score / final
Clean	mRNA tumor-suppressor delivery system	1.00	1.0 / 1.00
Dirty	(same doc, now ranked #2)	1.00	0.0 / 0.70

With clean IPC codes, the document with a perfect classification match gets pulled up to rank 1. This happens even though it does not have the single highest raw cosine score in the set. With dirty codes, that same document’s IPC term collapses to 0.0, since it has no overlap with the unrelated taxonomy. Its final score drops from 1.00 to 0.70, and a different document takes rank 1 instead. That other document has no IPC codes, so it falls back to cosine-only scoring. This confirms that the fusion term has a real, correctly-directed effect on the ranking. It is not just a no-op.

Part III

longdoc-rag/: chunked retrieval for long documents

9 Goal

Real patent applications and technical filings often run to thousands of words. This is far beyond BGE-small’s 512-token input limit. If I embed the whole document in one pass, one of two bad things happens. Either the document gets truncated, which silently throws away content past the limit, or many unrelated sections get blurred into a single vector that does

not represent any of them well. This pipeline solves that problem for long plain-text query documents. It splits a document into overlapping chunks, scores each chunk against the corpus on its own, and then combines the per-chunk scores into one ranked list per corpus document.

10 Chunking methodology

Tokenizer. I tokenize the document using BGE-small’s own tokenizer, accessed through `model.tokenizer`. I do not use a generic word splitter. Chunk boundaries need to be counted in the same units the model will actually truncate on.

Chunk size. Each chunk is 300 tokens long. This sits comfortably under BGE-small’s 512-token limit.

Overlap. Consecutive chunks share 15 percent of the chunk size. This way, a sentence or fact that falls on a chunk boundary still appears intact in at least one chunk. This overlap gives a **stride**, which is how far the window moves forward each step. The stride is roughly $0.85 \times 300 \approx 255$ tokens.

Sliding window. The first chunk covers tokens $[0, 300)$. The second chunk covers tokens $[255, 555)$. Each following chunk starts where the stride places it, and this continues until the document is covered.

Tail handling. If the tokens left over are fewer than the chunk size, I take them as-is for the final chunk. I do not pad them, and I stop there.

Per-chunk embedding. I treat each chunk as its own search query. I add the same BGE instruction prefix used elsewhere, embed the chunk with L2-normalized output, and score it against every normalized corpus document embedding using cosine similarity. Because the embeddings are normalized, this cosine similarity is just a dot product.

11 Aggregating chunk scores into one ranking

Chunking produces a score matrix with one row per chunk and one column per corpus document, not a single ranking. This matrix needs to be collapsed down to one score per document. The common default is plain max-pooling, where the single best chunk wins. But for very long, noisy documents, a single lucky chunk can produce a false positive. Instead, I use a **top- k average**. Here, k scales with how many chunks the query produced:

$$k = \text{clamp}(\text{round}(0.1 \times \text{num_chunks}), 3, 80)$$

In other words, I average the k best-scoring chunks for each document, where k is 10 percent of the total chunk count. I set a floor of 3 for k , so short documents still get some averaging instead of pure max-pooling. I also set a cap of 80, so very long documents do not dilute back toward a corpus-wide mean, which would wash out real signal. As a result, a document

needs several consistently strong-matching chunks to rank highly. A single spurious hit in an otherwise irrelevant document is not enough on its own.

12 End-to-end pipeline

`chunked_retrieval.py` reuses the corpus-indexing logic from `standard-rag/bge_retrieval.py` without any changes. It uses the same model and the same normalized embedding cache. It only replaces the query side of the pipeline:

1. Read the input as a plain-text file, with no JSON wrapper and no IPC metadata, so scoring uses default semantic weighting only.
2. Chunk the input following the methodology above.
3. Embed and score every chunk against the full corpus.
4. Aggregate the chunk scores with the top- k average to get one score per corpus document.
5. Rank the documents and return the top- k patents, using the same output schema (`rank`, `lens_id`, `title`, `abstract`, `ipc`, `score`) that the other two pipelines use, plus chunking metadata (`num_chunks`, `chunk_size_tokens`, `chunk_overlap_pct`, `k_used`) for reproducibility.

I ran a smoke test against a roughly 1,700-word synthetic patent describing a hydroponics apparatus. This produced 9 chunks and a k of 2. The pipeline correctly surfaced the most relevant corpus patents at the top of the ranking. These included greenhouse cultivation, tunable LED lighting systems, UV-sterilization electrode arrays, and bioreactor automation. Ranking quality dropped off gradually rather than sharply toward the tail of the top-10. That is what I would expect for a 7,392-document corpus at this similarity range.